SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Criteria	**Relevance to Application (1.25)**	**Use of Cloud Concepts (1.25)**	**Practical Insight (1)**	**Completeness (0.75)**	**Clarity (0.75)**	**TOTAL (5)**
Weightage	**25%**	**25%**	**20%**	**15%**	**15%**	**100%**
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty Total Marks Awarded:

Application Based Questions

1. How does cloud federation improve resource sharing and availability across datacenters?
2. What are the key security issues in virtualization, and how can they be mitigated?
3. How does a software-defined datacenter enhance scalability and management?
4. How can identity and access control ensure data privacy in cloud-based healthcare systems?
5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Q1. How does cloud federation improve resource sharing and availability across datacenters?

Answer:

Cloud federation refers to the interconnection of two or more independent cloud service providers (or geographically distributed datacenters of the same provider) so that they can share resources such as compute, storage, and network capacity through a common set of standards and interoperable APIs. Rather than each provider operating in isolation, federation allows workloads to move seamlessly across cloud boundaries based on availability, cost, latency, or regulatory requirements.

Key ways federation improves resource sharing and availability:

- Load balancing and burst capacity: When one datacenter reaches capacity, excess demand can be transparently "cloud-bursted" to a partner datacenter, avoiding service degradation.
- Geographic redundancy: Data and services replicated across federated sites ensure that a failure, outage, or natural disaster in one region does not bring down the entire application, directly improving availability (higher uptime/SLA compliance).
- Resource pooling: Federation creates a larger, shared pool of virtualized resources (VMs, storage, bandwidth) that can be dynamically allocated where they are needed most, improving overall utilization efficiency.
- Reduced latency: Requests can be routed to the nearest or least-loaded federated datacenter, improving response time for end users distributed across regions.
- Vendor independence: Federation reduces vendor lock-in by allowing organizations to distribute workloads across multiple providers, negotiating better pricing and avoiding single points of failure.

In practice, technologies such as identity federation (single sign-on across clouds), standardized APIs (e.g., OCCI, TOSCA), and inter-cloud brokers coordinate this resource sharing, making the federated system behave like a single, elastic, highly available cloud from the user's perspective.

Q2. What are the key security issues in virtualization, and how can they be mitigated?

Answer:

Virtualization is the foundation of cloud computing, but sharing physical hardware among multiple virtual machines (VMs) through a hypervisor introduces several unique security risks:

- VM Escape: A malicious process inside a guest VM breaks out of its isolation boundary and gains access to the hypervisor or host system, potentially compromising every VM on that host. Mitigation: keep the hypervisor patched, use hardware-assisted virtualization (Intel VT-x/AMD-V), and minimize the hypervisor's attack surface (thin/microkernel hypervisors).

- Hypervisor vulnerabilities: Since the hypervisor controls all VMs, any flaw in it is a single point of compromise for the entire host. Mitigation: regular security patching, vendor-supported hypervisors, and continuous vulnerability scanning.
- Data leakage / VM-to-VM attacks (side-channel attacks): Co-located VMs sharing CPU cache, memory, or storage can leak information through timing or cache-based side channels. Mitigation: workload isolation policies, cache partitioning, and avoiding co-residency of sensitive and untrusted workloads.
- Insecure VM images/templates: Cloning VMs from outdated or misconfigured images propagates vulnerabilities at scale. Mitigation: use verified, regularly updated golden images and automated image-scanning tools.
- Insufficient isolation of virtual networks: Misconfigured virtual switches or VLANs can allow traffic to cross tenant boundaries. Mitigation: micro-segmentation, virtual firewalls, and strict network policy enforcement.
- Malicious insiders / weak access control: Administrators with excessive privileges over the hypervisor can access or manipulate any tenant's VM. Mitigation: role-based access control (RBAC), multi-factor authentication, and detailed audit logging of all hypervisor-level actions.

Overall, a defense-in-depth approach — combining hypervisor hardening, strong isolation mechanisms, continuous monitoring, and strict identity and access management — is essential to secure virtualized cloud environments.

Q3. How does a software-defined datacenter enhance scalability and management?

Answer:

A Software-Defined Datacenter (SDDC) abstracts and virtualizes all infrastructure resources — compute, storage, and networking — and delivers them as software-controlled services, managed through a centralized automation and orchestration layer rather than manual, hardware-specific configuration.

Its three core pillars and their contribution to scalability and management are:

- Software-Defined Compute (SDC): Virtualization/hypervisors allow VMs and containers to be provisioned, resized, or migrated in seconds. This enables elastic, on-demand scaling of compute resources to match workload demand automatically.
- Software-Defined Storage (SDS): Physical storage devices are pooled and abstracted, so capacity can be added or reallocated without downtime, and storage policies (replication, tiering) can be applied programmatically per workload.
- Software-Defined Networking (SDN): Network control is decoupled from physical switches/routers, allowing network topology, load balancing, and security policies to be reconfigured through software (e.g., via APIs) instead of manual cabling or switch configuration.

Because every resource is exposed through APIs, an SDDC can be centrally orchestrated (e.g., using VMware vRealize, OpenStack, or Kubernetes-based platforms), enabling policy-driven automation: resources scale up or down automatically based on real-time metrics, provisioning that once took days can be completed in minutes, and a single management console gives administrators unified visibility and control over the entire datacenter. This significantly reduces operational overhead, human error, and time-to-deployment while allowing the datacenter to scale elastically to meet fluctuating cloud workloads.

Q4. How can identity and access control ensure data privacy in cloud-based healthcare systems?

Answer:

Healthcare systems store highly sensitive patient data (electronic health records, diagnostic images, billing information), making strict identity and access management (IAM) essential for privacy and regulatory compliance (e.g., HIPAA, GDPR).

Key IAM mechanisms and their role in protecting privacy:

- Authentication: Multi-factor authentication (MFA) verifies that only legitimate clinicians, staff, or patients can log in, preventing unauthorized access even if credentials are stolen.
- Role-Based Access Control (RBAC) / Attribute-Based Access Control (ABAC): Access to patient records is granted strictly according to job role (e.g., a nurse can view but not edit billing records; a doctor can access only their assigned patients' data), enforcing the principle of least privilege.
- Single Sign-On (SSO) and Federated Identity: Allows secure, centralized authentication across multiple healthcare applications and partner organizations without requiring each system to store duplicate credentials, reducing the attack surface.
- Encryption tied to access policies: Data is encrypted at rest and in transit, and decryption keys are released only to authenticated, authorized identities, ensuring that even a data breach does not expose readable patient information.
- Audit logging and monitoring: Every access request is logged and continuously monitored, enabling detection of anomalous behavior (e.g., a user accessing an unusually large number of records) and providing an audit trail required for compliance reporting.
- Consent and dynamic access policies: Patient-controlled consent mechanisms can dynamically grant or revoke access to specific parts of their record, ensuring data is shared only with authorized providers for a specific purpose and duration.

Together, these identity and access control measures ensure that only the right individuals can access the right data, for the right reason, at the right time — which is the foundation of data privacy in cloud-based healthcare systems.

Q5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Answer:

Virtualization is the technology that allows a single physical server to be partitioned into multiple isolated virtual machines (VMs), each running its own operating system and applications, using a hypervisor as the control layer. This abstraction is what makes cloud computing elastic and resource-efficient.

Contribution to scalability:

- Rapid provisioning: New VMs can be spun up from templates within minutes (or seconds for containers), allowing applications to scale out horizontally in response to increased demand.
- Elastic scaling (auto-scaling): Hypervisors work with cloud orchestration layers to automatically add or remove VM instances based on real-time load, ensuring performance during peak demand and cost savings during low demand.
- Live migration: Hypervisors (e.g., VMware ESXi, KVM, Xen) can move running VMs between physical hosts without downtime, allowing workloads to be redistributed for load balancing or hardware maintenance without interrupting service.

Contribution to resource efficiency:

- Server consolidation: Multiple VMs share the same physical hardware, increasing CPU, memory, and storage utilization instead of leaving servers underused, which reduces hardware and energy costs.
- Multi-tenancy: A single physical infrastructure can securely serve multiple customers/applications simultaneously, maximizing the return on physical infrastructure investment.
- Dynamic resource allocation: Hypervisors can dynamically adjust the CPU, memory, and I/O allocated to each VM based on real-time workload needs, avoiding both resource starvation and wastage.
- Fault isolation: Since each VM is isolated, a failure or crash in one VM does not affect others on the same host, improving overall system reliability without requiring dedicated hardware per application.

In summary, by decoupling applications from the underlying physical hardware, virtualization and hypervisors give cloud platforms the flexibility to scale resources up or down on demand while maximizing hardware utilization — the two properties that make cloud computing both elastic and cost-efficient.